

Optimization

Section 4.5

Steps for solving optimization problems

1. Information will be given in the problem to construct a ***constraint equation***. Usually this equation can easily be found by looking for the quantity (the actual number) given somewhere in the information. In most cases, there will be two or more variables in this equation, allowing you to solve for one of these.
2. Determine what quantity needs to be optimized (maximized or minimized). Write an equation for this quantity—this is your ***objective equation***. Your equation will again be one of at least two different variables.
3. Substitute the *constraint equation* into the *objective equation*. This will yield an objective equation of only one variable.
4. Once you have a single variable objective equation, it can be optimized by taking the derivative, setting it equal to zero and then solving for the variable.
5. Remember to answer the original question completely. You may need to plug the quantity into the objective equation to calculate a different quantity.

Hint: Make sure that units are kept constant—it is easier to equalize units at the beginning of a problem than during the procedure.

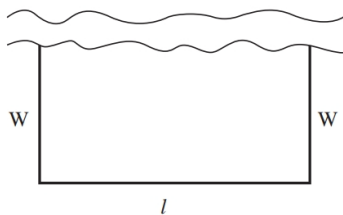
Refer to the page with geometrical formulas that was provided in Section 2.7. You are responsible for all of these formulas—they may not be provided to you during a test/exam.

Examples

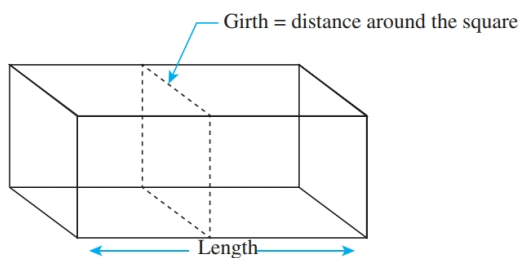
1. **(Video)** Given $f(x) = \frac{16x}{x^2 + 16}$, What is the maximum value of the function?

2. **(Video)** Find two numbers A and B (with $A \leq B$) whose sum is 56 and whose product is maximized.

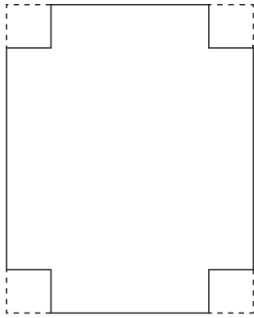
3. **(Video)** Farmer Brown has 6,000 ft of fence to create a rectangular pen that will be adjacent to a river. If he does not need to put any fence on the side that borders the river, what dimensions will maximize the area of the pen, and what is the maximum area?



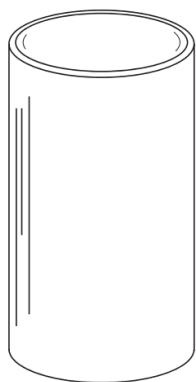
4. The U.S. Postal Service will accept a box for domestic shipping only if the sum of the length and the girth (distance around) does not exceed 108 inches. Find the dimensions of the largest volume box with a square end that can be sent.



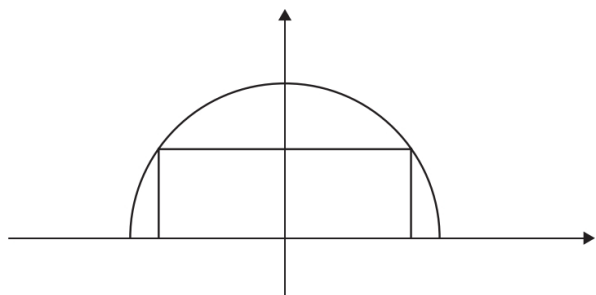
5. A box is to be made out of a 10 by 18 piece of cardboard. Squares of equal size will be cut out of each corner, and then the ends and sides will be folded up to form a box with an open top. Find the length L , width W , and height H of the resulting box that maximizes the volume (assume that $W \leq L$).



6. A cylindrical oatmeal container has a capacity of 3 L. Find the dimensions that will minimize the cost of production material to construct the container.



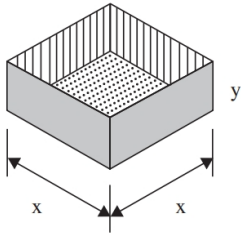
7. Find the area of the largest rectangle that can be inscribed in a semicircle with a radius 4.



8. Find the point on the line $y = 4x + 7$, which is closest to the point $(0, 0)$.



9. If 2,000 square centimeters of material is available to make a box with a square base and an open top, find the largest possible volume of the box.



10. A piece of wire 12 m long is cut into two pieces. One piece is bent into the shape of a circle of radius r and the other is bent into a square of side s . How should the wire be cut so that the total area enclosed is:

a. Maximized

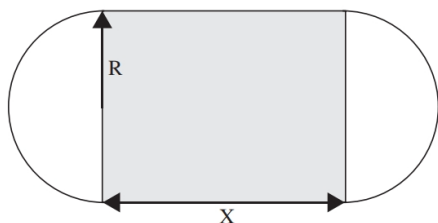
b. Minimized



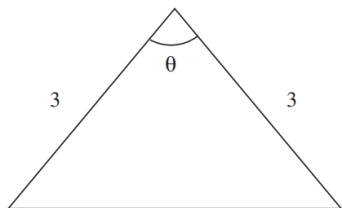
11. A Norman window has the shape of a semicircle atop a rectangle so that the diameter of the semicircle is equal to the width of the rectangle. What is the area of the largest possible Norman window with a perimeter of 45 ft?



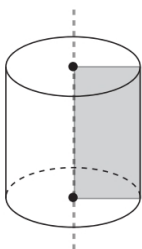
12. A running track has the shape of a rectangle with a semicircle on each end. If the length of the track is 400 m, find the dimensions so that
- The rectangular (shaded) region is maximized
 - The entire region is maximized



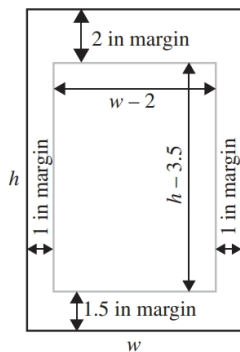
13. What angle θ between two edges of length 3 will result in an isosceles triangle with the largest area? (See diagram.)



14. Consider a rectangle of perimeter 12 in. Form a cylinder by revolving this rectangle about one of its edges. What dimensions of the rectangle will result in a cylinder of maximum volume?



15. A printer need to make a poster that will have a total area of 200 in.^2 and will have 1 in. margins on the sides, a 2-in. margin on the top and a 1.5-in. margin on the bottom. What dimensions will give the largest printed area?



16. There are 50 apple trees in an orchard. Each tree produces 800 apples. For each additional tree planted in the orchard, the output per tree drops by 10 apples. How many trees should be added to the existing orchard in order to maximize the total output of trees?